

Causality-Aware Preference Optimization for Aligning Medical Vision Language Models

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Abstract. Medical Large Vision-Language Models (Med-LVLMs) have substantially improved medical image understanding and reporting. However, existing models still suffer from modality bias, where correlations between background context and diagnostic labels cause over-reliance on spurious priors rather than lesion evidence, leading to clinically irrelevant or hallucinated outputs. To address this, we propose a **Causal Medical Preference Optimization (CaMedPO)** framework that learns an unbiased policy from a biased reference model. Unlike conventional direct preference optimization methods that inherit dataset biases, CaMedPO mitigates spurious causation while preserving the valuable indirect effect between lesion and background dependencies. This causal formulation yields a new preference-based loss with theoretical guarantees for bias mitigation and robust policy learning. The framework is model-agnostic and seamlessly integrates with existing Med-LVLM alignment pipelines. Extensive experiments on Med-VQA and medical report generation show that CaMedPO consistently improves factual accuracy and clinical relevance across multiple base models, significantly outperforming state-of-the-art DPO variants. Our results demonstrate that CaMedPO provides a principled route toward trustworthy, clinically grounded multimodal medical reasoning. Our code is available at <https://anonymous.4open.science/r/Med26E0>.

1 Introduction

The emergence of Large Vision Language Models (LVLMs) has advanced multimodal reasoning by enabling unified understanding of visual and textual modalities [?, ?, ?, ?]. Trained on large-scale image-text data, these models integrate strong visual perception with rich language knowledge and have shown impressive generalization across various tasks. In medicine, Medical LVLMs (Med-LVLMs) demonstrate promising capabilities in medical visual question answering [?, ?], radiology report generation [?, ?], and multimodal reasoning [?, ?], providing decision support and automated clinical documentation. However, these models still face the challenge of factuality, which mainly arises from modality misalignment between visual and textual streams [?, ?, ?]. Medical images contain subtle lesion features and imbalanced supervision across modalities, often leading Med-LVLMs to rely on textual priors instead of lesion-grounded evidence, resulting in hallucinated or clinically inconsistent responses [?].

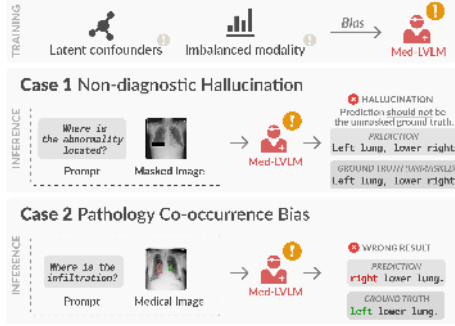


Fig. 1: Effects of spurious causation on inference. (a) Unexpectedly correct predictions given background image. (b) Co-occurrence of multiple pathologies affects the location prediction.

To address the factuality problem, recent works have introduced preference optimization to improve alignment between visual and textual modalities in Med-LVLMs. These methods fine-tune models by learning from pairwise preferences between clinically relevant and irrelevant responses, aiming to enhance factual grounding and reduce hallucinations. However, most existing approaches simply leverage the preference data generation process originally designed for aligning general LVLMs on natural images [?, ?, ?]. Directly transferring such pipelines to the medical domain neglects the unique characteristics of clinical data, where diagnostic correctness and lesion localization are essential. A recent study has highlighted the importance of clinical relevance in constructing effective preference data, proposing to evaluate and weight samples according to medical significance [?]. Nevertheless, these approaches still rely heavily on statistical associations between textual cues and disease labels, overlooking the influence of spurious causation that arises from biased modality interactions. As a result, even optimized models may continue to depend on spurious correlations rather than true lesion-grounded reasoning, limiting their factuality in clinical scenarios.

In medical reasoning, reliability depends strongly on the model’s ability to generate clinically consistent rationales rather than superficial correlations. However, existing Med-LVLMs often suffer from spurious causation, where predictions are influenced by latent confounders or imbalanced modality signals. One major cause arises from biased statistical associations, in which background regions or contextual patterns become spuriously linked to specific diseases. As illustrated in Fig. 1(a), even when an image contains only non-diagnostic background, the model can still predict the correct disease label, indicating an over-reliance on contextual priors instead of lesion evidence. Another common source of spurious effect is pathology co-occurrence bias, where correlated abnormalities lead the model to exploit coincidental features during learning. For example, as shown in Fig. 1(b), when pleural effusion and infiltration frequently appear together, the model tends to associate infiltration features with pleural effusion because the training data contain substantially more cases of infiltration in the

lower right lung and both conditions exhibit overlapping visual patterns. These observations highlight that mitigating spurious modality dependencies is essential for improving the factual reliability of Med-LVLMs in clinical reasoning.

A principled way to mitigate such spurious causation is to conduct a randomized controlled trial, collecting multimodal clinical data under diverse combinations of lesion and background conditions. In theory, such balanced sampling can disentangle visual evidence from contextual bias and reveal the true causal effect of lesion features on diagnostic outcomes [?]. However, this approach is infeasible in practice because it requires extensive manual annotation and controlled patient recruitment. Recent work such as MMedPO [?] partially addresses this issue by introducing local noise to lesion areas, which can be interpreted as an intervention on biased training distributions. Yet, this strategy remains sub-optimal since it fails to decouple beneficial background-lesion interactions from harmful ones. In real clinical images, background cues such as tissue density, lung boundary, or rib contrast often provide useful priors when jointly considered with lesion morphology [?, ?]. For example, distinguishing pleural effusion from lower-lobe infiltration depends not only on lesion opacity but also on surrounding pleural line continuity and diaphragm curvature [?]. Therefore, an effective debiasing framework should preserve clinically meaningful lesion-background relationships while removing confounding dependencies that distort modality alignment.

To address the above challenge, we propose a **Causal Medical Preference Optimization** framework (**CaMedPO**), which mitigates spurious causation while preserving the valuable indirect effect between lesion and background features. CaMedPO can be seamlessly integrated with existing DPO-based alignment methods to further enhance the performance of Med-LVLMs, serving as a general and model-agnostic optimization paradigm. Specifically, we formulate a causal graph that connects multimodal inputs to the reward function, enabling theoretical analysis of how biased reference policies influence downstream alignment. Based on this formulation, CaMedPO derives a principled objective that improves robustness even when the SFT-based reference model is biased. Within this causal structure, we explicitly decouple the harmful direct effect, which transmits bias from shortcut correlations, and the valuable indirect effect, which captures the clinically meaningful joint influence of lesion and contextual cues on diagnostic reasoning. During preference construction, we build factual and counterfactual outcome pairs by employing a lesion detection model to identify clinically relevant regions, which are then used to form paired samples for causal effect estimation. The factual outcomes preserve the normal lesion-background association, while the counterfactual ones intervene on lesion evidence to isolate indirect effects. In the inference phase, the model is trained on normal preference pairs, ensuring consistency with standard alignment while benefiting from causal debiasing during optimization. In summary, our main contributions are threefold:

- We introduce a causal framework to analyze how biased reference policies propagate harmful direct effects in medical vision-language alignment. This for-

116 mutation reveals that factuality degradation originates from spurious causation 116
 117 within modality interactions. 117

- 118 • We propose CaMedPO, a model-agnostic debiasing approach that mitigates 118
 119 spurious causal effects while preserving the valuable indirect effect between lesion 119
 120 and background features. CaMedPO constructs factual-counterfactual outcome 120
 121 pairs based on lesion detection and can be seamlessly integrated with existing 121
 122 DPO methods to enhance robustness. 122
- 123 • Extensive experiments on Med-VQA and medical report generation bench- 123
 124 marks demonstrate that CaMedPO consistently improves factual accuracy and 124
 125 clinical relevance across multiple Med-LVLMs, highlighting its broad applicabil- 125
 126 ity and effectiveness in trustworthy multimodal medical reasoning. 126

127 2 Related Works 127

128 **Preference Optimization for Med-LVLMs.** Preference optimization is widely 128
 129 adopted to align large vision-language models (LVLMs) with human or model 129
 130 preferences [?, ?, ?, ?, ?, ?, ?, ?]. More details are discussed in Appendix S6. Ex- 130
 131 tending such techniques to the medical domain introduces additional challenges, 131
 132 as factual grounding and clinical relevance are essential. Methods such as [?, ?, ?] 132
 133 generate dispreferred responses via GPT-4 or the Med-LVLM for preference opti- 133
 134 mization. Medical alignment emphasizes clinical grounding and lesion-aware sig- 134
 135 nals. MMedPO [?] proposes clinically aware weighting to preference samples to 135
 136 highlight key medical cues, while MedGR² [?] uses generative reward learning for 136
 137 robust medical reasoning. Despite these advances, transferring general-domain 137
 138 pipelines still leaves gaps: models over-rely on textual priors and spurious back- 138
 139 ground associations under imbalanced modality interactions [?, ?]. MMedPO at- 139
 140 tempts to address these issues by weighting samples based on clinical relevance, 140
 141 but relies on computationally heavy multi-agent scoring and does not preserve 141
 142 useful lesion-background priors. Guided by these observations, our work intro- 142
 143 duces a causally aware preference construction to probe lesion-mediated visual 143
 144 evidence while preserving clinically meaningful multimodal cues. 144

145 **Causal Reference.** Causal inference [?] uncovers cause-effect mechanisms through 145
 146 intervention and counterfactual reasoning. Interventions modify distributions for 146
 147 unbiased estimates, while counterfactuals imagine hypothetical outcomes to for 147
 148 alternative conditions. Recent learning-based approaches incorporate causal rea- 148
 149 soning into deep models to reduce spurious shortcuts and improve generalization 149
 150 across vision tasks [?, ?, ?], including anomaly detection [?], visual recognition [?], 150
 151 recommendation [?, ?], and scene graph generation [?]. In multimodal models, 151
 152 causal modeling is explored to improve robustness and interpretability. Wang 152
 153 *et al.* [?] propose causal reward modeling to prevent reward hacking, but their 153
 154 framework is limited to unimodal text and does not extend to medical multi- 154
 155 modal reasoning. Zhou *et al.* [?] integrate structural causal models into medical 155
 156 VQA to interpret visual-text relations, yet their front-door adjustment requires 156
 157 heavy sampling and fails to preserve lesion-background dependencies, rendering 157
 158 it suboptimal for multimodal alignment. Building on these insights, we extend 158

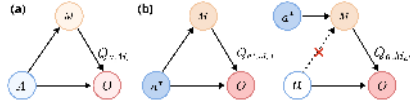


Fig. 2: (a) Example of causal graph. (b) Examples of counterfactual notations.

causal counterfactual reference to preference optimization. By constructing factual and counterfactual pairs, our framework suppresses harmful direct effects from biased supervision while preserving the valuable indirect effect that capture clinically meaningful lesion–background dependencies.

3 Preliminaries

We introduce the core notation and concepts used in our causal preference framework: the Med-LVLM formulation, preference optimization, and causal counterfactual learning.

A Med-LVLM combines a visual encoder with an LLM to process multimodal medical inputs. Given a medical image I and a clinical query T , the input is $x = (I, T)$, and the model autoregressively predicts next token probabilities $\pi_\theta(y_i|y_{<i}, x)$, yielding sequence likelihood $\pi_\theta(y|x) = \prod_i \pi_\theta(y_i|y_{<i}, x)$, where y is the generated medical report or answer.

Preference Optimization. Preference optimization aligns the conditional policy using pairwise preference supervision, x, y_w, y_l . DPO [?] directly connects policy with the reward difference between preferred and dispreferred responses. To remove the explicit reward model, DPO reformulates this closed-form expression as a supervised learning objective. Given a preference dataset $\mathcal{D} = \{(x^{(i)}, y_w^{(i)}, y_l^{(i)})\}_{i=1}^N$, where y_w and y_l denote the preferred and dispreferred responses for input x , the DPO loss is

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\eta^{-1} \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \eta^{-1} \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]. \quad (1)$$

We provide the complete derivation in Appendix S1.

Causal Counterfactual Learning. Causal counterfactual learning formalizes how outcomes change under hypothetical interventions using structural causal models (SCMs) [?]. We define a causal graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of variables and directed edges, respectively. As shown in Fig. 2, for a causal chain $A \rightarrow M \rightarrow O$ A denotes the *cause*, M represents the *mediator*, and O is the *effect* or *outcome*. The counterfactual outcome, $O_{a,m}$ represents the value that O would take if A were set to a and M were set to m , i.e., $O_{a,m} = O(A=a, M=m)$.

In the factual world, $M = M_a$ and the observed outcome is B_{a,M_a} . In the counterfactual world, A may take a different hypothetical value a^* , while the

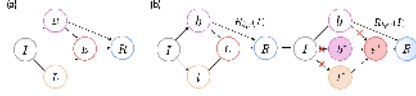


Fig. 3: Comparison between (a) SCM of Med-LVLM and (b) Counterfactual SCM of Med-LVLM.

mediator is fixed to its factual value M_a , yielding O_{a^*, M_a} . Causal effects quantify the expected outcome difference between two treatment settings $A = a$ and $A = a^*$. According to the causal theory [?], the Total Effect (TE) of A on B is defined as

$$\text{TE} = O_{a, M_a} - O_{a^*, M_{a^*}}, \quad (2)$$

which decomposes into the Natural Direct Effect (NDE)

$$\text{NDE} = O_{a, M_{a^*}} - O_{a^*, M_{a^*}}, \quad (3)$$

and the Total Indirect Effect (TIE)

$$\text{TIE} = \text{TE} - \text{NDE} = O_{a, M_a} - O_{a, M_{a^*}}. \quad (4)$$

where M_{a^*} denotes the value of M had A been a^* [?]. NDE isolates the direct causal path from A to O while blocking M , whereas TIE captures the mediated contribution through M .

4 Methodology

4.1 Cause-Effect Look in Preference Optimization

Causal Analysis As illustrated in Fig. 3, the proposed causal graph for medical multimodal alignment consists of four key components: the medical image input I , its decomposed visual factors including the background B and lesion evidence L , the learned joint features E , and the resulting reward model R . Based on the causal dependencies among these variables, we summarize the major links as follows.

Link $I \rightarrow B \rightarrow R$ captures the harmful shortcut between the original input image and the reward through biased background correlations. Background factors B (e.g., scanner type, imaging device, or patient posture) may be spuriously correlated with certain labels or textual descriptions, leading the model to achieve high reward even when clinically irrelevant regions dominate the decision [?].

Link $B \leftarrow I \rightarrow L$ depicts the generation of both background and lesion representations from the same medical image input. I is processed by the visual encoder to produce background features B and lesion features L together. B corresponds to global background such as lung field structure or imaging device signatures, while L represents suspicious regions detected by attention or segmentation models.

Link $B/L \rightarrow E \rightarrow R$. captures the total causal influence of visual representations on the reward through the embedding E , which aggregates both background and lesion features learned from the visual encoder. In particular, B provides the valuable contextual prior along the good causal link $B \rightarrow E \rightarrow R$, which facilitates reliable recognition when lesion information are subtle or visually confounded by surrounding anatomical structures. For example, in chest radiography, global contextual cues such as lung symmetry, cardiac silhouette, or diaphragm curvature help disambiguate faint opacities and improve diagnostic localization [?]. This clinically grounded pathway allows CaMedPO to preserve meaningful joint information while mitigating harmful direct effects from B on R .

Causal Reward To isolate clinically meaningful signals from biased supervision, we model the causal effect of the background factor B on the reward R through the ensemble mediator E . Specifically, the factual reward generation process under a given input image I can be expressed as:

$$R_{b,e}(I) = R(B = b, E_{b,l} = E(B = b, L = l) \mid I). \quad (5)$$

The factual reward $R_{b,e}(I)$ inherently mixes both the useful lesion-grounded information and the harmful direct effect from B . To quantify these distinct contributions, we first define the Total Effect (TE) as:

$$\text{TE} = R_{b,e}(I) - R_{b^*,e^*}(I), \quad (6)$$

where b^* and e^* correspond to the counterfactual background and its associated ensemble embedding. We next estimate the Natural Direct Effect (NDE) of B for the pure harmful background bias:

$$\text{NDE} = R_{b,e^*}(I) - R_{b^*,e^*}(I). \quad (7)$$

$R_{b,e^*}(I)$ denotes the counterfactual outcome:

$$R_{b,e^*}(I) = R(B = b, E_{b^*,l^*} = E(B = b^*, L = l^*) \mid I). \quad (8)$$

As the indirect effect of embedding E on the link $I \rightarrow B/L \rightarrow E \rightarrow R$ is blocked, the reward model can only be biased relying on the direct effect in the link $I \rightarrow B \rightarrow R$. To mitigate the background bias in NDE, we compute the Total Indirect Effect (TIE) by subtracting NDE from TE:

$$\text{TIE} = R_{b,e}(I) - R_{b,e^*}(I). \quad (9)$$

Intuitively, TIE represents the reward improvement driven by medically relevant background-lesion interactions.

4.2 Theoretical Analysis

In standard DPO, the optimization is based on the relative preference difference between a preferred sample y_w and a dispreferred one y_l , where the reward

difference $r(x, y_w) - r(x, y_l)$ determines the training direction. However, such rewards may still encode background-related biases that affect both positive and negative samples. To obtain a bias-free causal preference, we define the conventional reward r with the total indirect effect r_{TIE} as expressed in Eq. 9. Substituting r_{TIE} into the Bradley–Terry model, the preference probability between (y_w, y_l) becomes

$$P(y_w \succ y_l \mid x) = \sigma(\eta[r_c(x, y_w) - r_c(x, y_l)]), \quad (10)$$

where $\sigma(\cdot)$ is the logistic function. The difference $r_c(x, y_w) - r_c(x, y_l)$ serves as a *causal contrast* that reflects the pure clinical contribution through the mediator E . Expanding this difference yields

$$\begin{aligned} r_c(x, y_w) - r_c(x, y_l) = & \underbrace{[R_{b,e}(x, y_w) - R_{b,e}(x, y_l)]}_{\text{factual difference}} \\ & - \underbrace{[R_{b,e^*}(x, y_w) - R_{b,e^*}(x, y_l)]}_{\text{counterfactual difference}}. \end{aligned} \quad (11)$$

This symmetric structure indicates that both the positive and negative samples are corrected by the same causal mechanism. If the correction were applied only to the preferred response y_w , the residual background effect on y_l would remain, and the pairwise reward difference would still contain the direct bias from B .

In preference optimization, the reference policy π_{ref} is typically obtained from a supervised fine-tuned (SFT) model that aligns to instruction-following data. However, π_{ref} may carry bias from the underlying training distribution, causing modality shortcuts in medical preference learning. Our goal is to show that the proposed optimization goal can theoretically correct these biases and converge to a causal-optimal policy. Formally, we define the causal-optimal policy and provide a convergence guarantee based on the unbiased TIE reward. Let π_{ref} be an arbitrary reference policy. The observed reward can be decomposed as

$$r_o(x, y) = r_c(x, y) + \delta(x, y),$$

where r_c is the causal TIE reward and δ denotes the bias caused by the direct background effect. Assume all rewards are bounded by $|r_c|, |\delta| \leq R_{\text{max}}$.

Definition 1 (Causal-optimal policy). *Given a biased reference policy π_{ref} and the unbiased TIE reward $r_c(x, y) = R_{b,e}(I) - R_{b,e^*}(I)$, the causal-optimal policy is defined as*

$$\pi_c^*(y \mid x) \propto \pi_{\text{ref}}(y \mid x) \exp(\eta r_c(x, y)).$$

Theorem 1 (Convergence of causal reward). *Under Assumption S2 that all rewards are bounded by R_{max} , the bias magnitude satisfies $\|\delta\|_{\infty} \leq \varepsilon$, and the learning rate η is sufficiently small, the policy sequence updated by TIE-DPO satisfies*

$$\begin{aligned} \mathbb{E}[\text{KL}(\pi_{T+1} \parallel \pi_c^*)] & \leq \text{KL}(\pi_1 \parallel \pi_c^*) - \eta \sum_{t=1}^T \mathbb{E}_{x,e}[\text{Var}_{\pi_t}(r_c)] \\ & + O(T\eta^2 R_{\text{max}}^2). \end{aligned}$$

which ensures a monotonic improvement toward the causal-optimal policy π_c^* .

The bias term δ amplifies spurious correlations between background and reward, causing the exponentiated policy update to deviate from the causal optimum. Full derivations and proofs are provided in Appendix S2.

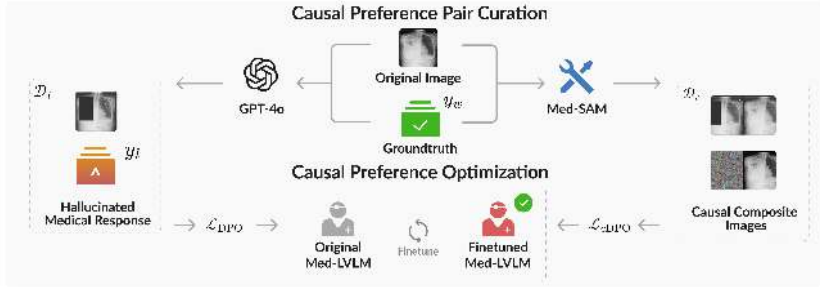


Fig. 4: The overview of CaMedPO that consists of the causal preference pair curation and causal preference optimization.

4.3 Implementation Instantiation

Causal Preference Pair Curation To realize the causal preference structure defined in Eq. 11, we curate paired data that reflect both factual and counter-factual medical reasoning, where the whole framework is illustrated in Fig. 4. The construction proceeds in two stages.

Generating Hallucinated Medical Responses. Following the causal contrastive formulation, we first obtain dispreference responses y_i by introducing factual inconsistencies into medical reports, following the approach in MMedPO [?]. Specifically, we employ GPT-4o [?] to generate multiple candidate responses for each case sampled from the target Med-LVLMs, and evaluate their factual consistency against the ground-truth response y_w . The candidate exhibiting the highest hallucination score, such as incorrect image interpretation or misleading diagnostic reasoning, is selected as the dispreference response. When no candidate response exhibits sufficient deviation, GPT-4o is prompted to synthesize a dispreference response. The preference pairs constructed by this process are denoted as $\mathcal{D}_t$.

Constructing Causal Composite Images. To explicitly instantiate the causal mechanism illustrated in our causal graph, we construct composite visual inputs that allow controlled intervention on lesion and background evidence. Specifically, we employ a medical visual localization tool to segment the lesion area from the original image I , identifying the regions most relevant to disease diagnosis. The lesion mask is then removed, and the masked region is filled with zero-valued pixels to form a background-only image I_b , representing the

background variable $B=b$ in the structural causal model. The original image I serves as the factual visual condition associated with the denoted $E=e$, while a Gaussian noise-perturbed version of the same image serves as the counterfactual condition $E=e^*$.

These components are combined into composite image pairs that preserve the same anatomical context but differ in the lesion evidence, forming the factual and counterfactual visual conditions: $(I_b \oplus I)$ and $(I_b \oplus I_{\text{null}})$, where $\oplus$ denotes image stitching. Each pair is associated with a clinical query prepended by an instruction emphasizing that both images come from the same patient, such as “*Both images belong to the same patient. Please analyze each and then provide a joint clinical conclusion*”, where the revised clinical query is denoted as T' . The preference data constructed in this stage is denoted as $\mathcal{D}_c$. By aligning the composite inputs with their preferred and dispreference textual responses, we obtain the overall preference dataset $\mathcal{D}_f = \mathcal{D}_t \cup \mathcal{D}_c = \{(x^{(i)}, x^{*(i)}, x_c^{(i)}, x_c^{*(i)}, y_w^{(i)}, y_l^{(i)})\}_{i=1}^N$, where $x^{(i)} = (I, T)$ and $x^{*(i)} = (I_b, T)$ are the standard preference pairs used in MMedPO, while $x_c^{(i)} = (I_b \oplus I, T')$ and $x_c^{*(i)} = (I_b \oplus I_{\text{null}}, T')$ are the causal pairs.

Causal Preference Optimization Changing R with its logarithmic likelihood form used in DPO, we obtain the causal logit

$$\begin{aligned} \mathcal{A}_{\text{CDPO}} = & \left[\log \frac{\pi_{\theta}(y_w | x_c)}{\pi_{\text{ref}}(y_w | x_c)} - \log \frac{\pi_{\theta}(y_w | x_c^*)}{\pi_{\text{ref}}(y_w | x_c^*)} \right] \\ & - \left[\log \frac{\pi_{\theta}(y_l | x_c)}{\pi_{\text{ref}}(y_l | x_c)} - \log \frac{\pi_{\theta}(y_l | x_c^*)}{\pi_{\text{ref}}(y_l | x_c^*)} \right]. \end{aligned} \quad (12)$$

Therefore, the causal optimization objective is

$$\mathcal{L}_{\text{CDPO}} = -\mathbb{E}_{(x_c, x_c^*, y_w, y_l) \sim \mathcal{D}_f} [\log \sigma(\eta^{-1} \mathcal{A}_{\text{CDPO}})], \quad (13)$$

Note that the original MMedpo preference optimization goal is also retrained to keep the Med-LVLM has the ability to make predictions according to the original visual and clinical query inputs. The optimization objective is

$$\begin{aligned} \mathcal{L}_{\text{DPO}} = & -\mathbb{E}_{(x, x^*, y_w, y_l) \sim \mathcal{D}_f} \\ & \left[\log \sigma \left(\eta^{-1} \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \eta^{-1} \log \frac{\pi_{\theta}(y_l | x^*)}{\pi_{\text{ref}}(y_l | x^*)} \right) \right]. \end{aligned} \quad (14)$$

However, we do not employ the clinical relevance score used in MMedPO to reduce the computational workload. Finally, the objective of CaMedPO is the combination of the two preference optimizations:

$$\mathcal{L}_{\text{CaMedPO}} = \mathcal{L}_{\text{DPO}} + \mathcal{L}_{\text{CDPO}}. \quad (15)$$

5 Experiments

5.1 Experimental Setups

Evaluation Datasets. We evaluate on the Med-VQA benchmarks VQA-RAD [?] and SLAKE [?], and on the radiology report benchmarks IU X-Ray [?] and

Table 1: Performance comparison on medical VQA and report generation tasks covering SLAKE, VQA-RAD, and IU-XRay datasets. For open-ended questions, we report recall (Open); for closed-ended questions, accuracy (Closed). The BLEU score denotes the average of BLEU-1/2/3/4. +SFT indicates that the model is first fine-tuned with SFT before applying the corresponding baselines. The best and second-best results are highlighted in **bold** and underlined, respectively.

2* Models	SLAKE		VQA-RAD		IU-XRay			MIMIC-CXR		
	Open	Closed	Open	Closed	BLEU	ROUGE-L	METEOR	BLEU	ROUGE-L	METEOR
LLaVA-Med v1.5	47.54	52.99	29.50	64.54	14.56	10.31	10.95	10.25	9.38	7.71
+ STLLaVA-Med	50.48	54.67	35.00	68.53	16.10	10.58	10.51	11.11	9.29	7.72
+ DPO	49.84	53.59	32.00	66.53	16.08	12.95	17.13	11.19	9.45	7.80
+ mDPO	51.20	60.29	35.00	68.13	18.43	22.38	26.52	11.55	9.62	7.90
+ SPPO	52.62	67.58	38.50	68.93	21.66	27.30	32.10	12.32	10.19	8.83
+ MMedPO	51.67	58.25	36.50	67.33	<u>23.49</u>	29.52	<u>34.16</u>	12.85	<u>11.13</u>	<u>10.03</u>
cyan!7 + CaMedPO w/ mDPO	51.67	65.07	38.50	68.53	22.67	<u>30.65</u>	33.24	12.31	10.58	9.88
cyan!7 + CaMedPO w/ SPPO	<u>53.73</u>	<u>69.02</u>	42.50	<u>69.72</u>	21.46	27.17	32.23	12.54	10.56	9.78
cyan!7 + CaMedPO	53.11	71.17	<u>40.00</u>	71.31	23.78	32.68	35.26	<u>12.65</u>	11.61	10.37
+ SFT	52.62	58.01	31.50	64.14	22.75	28.86	33.66	12.39	10.21	8.75
+ STLLaVA-Med	54.53	61.72	33.50	70.12	22.97	28.98	34.05	12.21	10.12	8.98
+ DPO	53.49	64.23	33.00	64.94	23.82	29.97	34.89	12.37	10.38	9.10
+ mDPO	53.10	63.88	34.50	67.33	23.57	28.35	33.13	11.77	9.45	8.91
+ SPPO	58.57	66.99	41.00	73.31	23.78	30.05	34.78	12.87	12.32	9.45
+ MMedPO	56.60	67.82	37.00	67.73	24.00	30.13	35.17	<u>13.28</u>	<u>13.22</u>	10.20
cyan!7 + CaMedPO w/ mDPO	57.56	69.02	40.00	71.31	25.28	33.69	37.42	12.79	11.35	10.04
cyan!7 + CaMedPO w/ SPPO	<u>61.66</u>	<u>70.93</u>	44.00	<u>74.10</u>	<u>27.33</u>	<u>34.38</u>	<u>38.11</u>	12.84	12.60	<u>11.63</u>
cyan!7 + CaMedPO	62.17	74.40	<u>42.50</u>	74.90	26.23	34.13	38.25	14.91	14.58	12.22

MIMIC-CXR [?]. We adhere to official splits and standard preprocessing to ensure strict comparability; dataset statistics are provided in Appendix S3.

Implementation Details. We use LLaVA-Med-1.5-mistral-7B [?] as the base model. Med-SAM [?] with clinical query guidance is used to localize the lesion area. The visual encoder is CLIP ViT-L14 at input resolution 336. Optimization employs LoRA with rank 128 and scaling 256. Dropout is set to 0.05 and no LoRA bias is used. Training runs for three epochs with a per-device batch size of two and gradient accumulation of one. We set the learning rate to 1e-6 with cosine scheduling and a warmup ratio of 0.03. Weight decay is zero. We save at the end of each epoch, keeping a single retained checkpoint. Prompts and context formatting are kept consistent across methods. All experiments are conducted on 8 NVIDIA A100 GPUs.

Baselines. We compare CaMedPO with DPO [?] and its variants on medical tasks, including STLLaVA-Med [?] and MMedPO [?]. Note that CaMedPO can be integrated with other DPO methods for improvement by modifying $\mathcal{L}_{\text{DPO}}$ in Eq. 15. We further compare our method against two preference optimization methods for natural image. Specifically, mDPO [?] serves as a direct preference optimization method, while SPPO [?] is a self-rewarding model. We integrate our causal preference optimization framework with both methods and use them as baselines to assess comparative performance. In addition, we also evaluate MMedPO and all baseline models after SFT. Further implementation details are provided in Appendix S4.

Evaluation Metrics. For VQA, we report normalized exact-match accuracy, where clinical correctness and hallucination are assessed with the Lingshu [?] evaluation framework MedEvalKit. For report generation, we report BLEU Score [?], ROUGE-L [?], and METEOR [?].

5.2 Main Results

We note that the SLAKE and VQA-RAD test sets used in MMedPO differ from the official splits. For fair comparison, we therefore re-evaluate LLaVA-Med [?], STLLaVA-Med [?], DPO [?], and MMedPO [?] using the Lingshu toolkit, while official results are retained for IU-XRay and MIMIC-CXR. CaMedPO is introduced as a causal extension of the unweighted MMedPO and is further applied to mDPO [?] and SPPO [?] to assess the generality of the proposed causal preference optimization framework.

Comparison with Baseline Methods. As reported in Table 1, CaMedPO achieves consistent improvements over all baseline preference optimization methods across both medical VQA and report-generation tasks. When compared with MMedPO [?], CaMedPO exhibits clear gains on multiple benchmarks. For example, on SLAKE, CaMedPO improves open-ended recall from 51.67% to 53.11% and closed-ended accuracy from 58.31% to 71.16%. On IU-XRay, CaMedPO also yields consistent gains, improving ROUGE-L from 29.52% to 32.68%. These representative improvements indicate that CaMedPO provides a more reliable preference signal, yielding better alignment of medical reasoning across diverse datasets.

Comparison with Baseline Methods Enhanced by SFT. As shown in Table 1, CaMedPO continues to outperform all SFT-enhanced baselines, with more improvements than in the non-SFT setting. Relative to MMedPO [?] after supervised fine-tuning, CaMedPO achieves larger gains across both VQA and report-generation benchmarks. For example, on SLAKE-Closed, performance increases from 67.82% to 74.40%, and on VQA-RAD closed-ended accuracy from 67.73% to 74.90%. On IU-XRay, CaMedPO further improves ROUGE-L from 30.13% to 34.13%. These results suggest that SFT provides a stronger reference policy that amplifies the benefits of causal preference optimization. By decoupling background-related bias from the preference signal, CaMedPO is able to extract more reliable supervision from the improved SFT initialization.

Applicability to Different Preference Optimization Algorithms. Beyond enhancing MMedPO, the proposed causal adjustment is also applied to mDPO and SPPO to assess its generality. In both cases, the causal variants exhibit consistent improvements over their baselines. For mDPO [?], CaMedPO increases SLAKE closed-ended accuracy from 60.29% to 65.07% and improves IU-XRay ROUGE-L from 22.38% to 30.65%. For SPPO, CaMedPO further raises VQA-RAD open-ended recall from 38.50% to 41.00% and achieves stronger report-generation results, using the second-round SPPO outputs for evaluation. These observations indicate that the proposed causal correction provides a transferable enhancement that strengthens preference optimization algorithms.

Table 2: Ablation study results on VQA-RAD and IU-XRay. “w/” and “w/o” denote with and without, respectively. ^s denotes the model enhanced with SFT.

2*Methods	VQA-RAD		IU-XRay		
	Open	Closed	BLEU	ROUGE-L	METEOR
MMedPO ^s	37.00	67.73	24.00	30.13	35.17
CaMedPO ^s	42.50	74.90	26.23	34.13	38.25
CaMedPO ^s w/o $\mathcal{L}_{\text{DPO}}$	29.50	57.77	14.11	10.65	10.37
CaMedPO ^s w/ causal-weighted $\mathcal{L}_{\text{DPO}}$	37.50	70.52	24.25	28.40	34.64
CaMedPO ^s w/ counterfactual query	42.00	72.91	26.56	34.02	36.68
CaMedPO ^s w/ MedKLIP	38.50	73.31	25.62	32.07	37.09
MMedPO ^s w/ background	26.50	65.34	15.77	10.36	10.05
CaMedPO ^s w/ background	22.00	53.78	13.29	9.45	10.58
CaMedPO ^s w/ composite input	37.50	69.32	22.35	28.69	32.57

5.3 Ablation Study

In Table 2, we select SFT-enhanced MMedPO as the CaMedPO baselines to conduct ablation studies on VQA-RAD and IU-XRay datasets. Multi-rounds training, sample ratio of causal pairs, and compatibility analysis on other LVLMS are in Appendix S5.

Loss Function Variants. We evaluate several loss function variants to understand their contribution. Removing the standard DPO term leads to a clear performance drop (for example, VQA-RAD open-ended recall decreases from 42.00% to 29.50%), indicating that learning from factual preference pairs remains essential. Applying causal weights to the original DPO loss, where the details is in Appendix S5 improves over MMedPO but remains weaker than the full model, as reflected by the IU-XRay ROUGE-L score of 28.40% compared with 34.13%. Removing the counterfactual clinical query provides only marginal improvements, thus it is not adopted in our causal preference optimization.

Medical Visual Localization Tool. We further examine the impact of the lesion-background decomposition module by replacing the Med-SAM [?] localization with MedKLIP [?]. CaMedPO^s w/ MedKLIP leads to a moderate performance drop, where VQA-RAD open-ended recall decreases by 4%, indicating that accurate lesion localization contributes to stronger causal contrast. Nevertheless, the model still outperforms the original MMedPO^s baseline in all metrics, showing that even imperfect localization retains the advantage of causal preference construction.

Reference Inputs Comparison. We further evaluate the effect of different visual inputs used for causal preference construction. When replacing the original image with the background-only image, MMedPO^s w/ background achieves higher performance than CaMedPO^s w/ background (e.g., VQA-RAD Closed Accuracy: 65.34% vs. 53.78%), suggesting that CaMedPO effectively suppresses

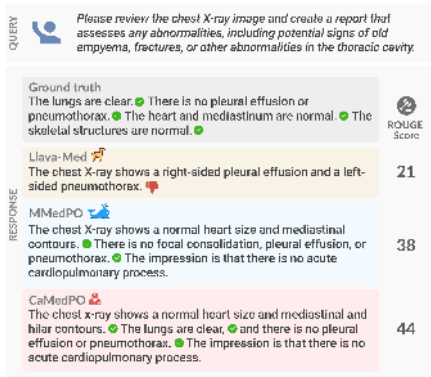


Fig. 5: CaMedPO generates reasonable responses.

the background-induced bias that MMedPO continues to exploit. We additionally test using the composite image as the sole visual input. While this variant performs better than background-only inputs, its performance remains lower than the full CaMedPO model, likely because the presence of explicit background channels still introduces confounding information. These observations are consistent with our causal graph formulation, where background can be the biased term.

6 Qualitative Analysis

To qualitatively assess the clinical reliability of CaMedPO, we examine representative examples comparing its outputs with those produced by LLaVA-Med and MMedPO. More cases are in Appendix S5. As illustrated in Fig. 5, LLaVA-Med produces an erroneous description, incorrectly suggesting bilateral pleural effusions in the report generation task. MMedPO improves factual accuracy but still omits clinically relevant details and exhibits partially incomplete reasoning. In contrast, CaMedPO provides a more comprehensive and clinically aligned report. It correctly identifies normal cardiac size, mediastinal contours, and clear lung fields, fully matching the ground-truth interpretation and achieving the highest ROUGE score.

7 Conclusion

This work presents CaMedPO, a causality-aware preference optimization framework that reduces background-induced shortcuts in medical large vision language model. Using counterfactual reasoning and isolating background-related evidence, CaMedPO produces a more reliable preference reward and yields consistent improvements in both medical VQA and report-generation tasks. Theoretical analysis further shows that CaMedPO converges closer to the causal-optimal policy even when the reference model is biased, reinforcing its validity

as a principled approach for medical alignment. Our findings also indicate that accurate lesion localization contributes meaningfully to downstream clinical reasoning. A natural direction for future research is to integrate lesion grounding more tightly with medical preference learning, for example through joint optimization strategies or cooperative game design that strengthen the alignment between visual evidence and clinical decision-making.

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S1 Detailed Preliminaries

Building upon the probabilistic formulation introduced above, preference optimization aims to align the conditional policy $\pi_\theta(y|x)$ with desired responses based on pairwise preference supervision. This paradigm originates from reinforcement learning from human feedback (RLHF) [?], where a reward model $r(x, y)$ is learned to reflect human preference and the policy is optimized with reinforcement learning algorithms such as PPO [?]. Following the Bradley–Terry model [?], the probability of preferring one response y over another y' given the same input x is modeled as $P(y \succ y'|x) = \sigma(r(x, y) - r(x, y'))$, where $\sigma(\cdot)$ denotes the sigmoid function. Given the true reward model $r(x, y)$, RLHF seeks to maximize the expected reward while constraining deviation from a reference policy π_{ref} , formulated as $\max_\theta \mathbb{E}_{x, y \sim \pi_\theta} [r(x, y)] - \eta^{-1} \mathbb{E}_x [\text{KL}(\pi_\theta(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x))]$, where η controls the trade-off between reward maximization and policy regularization.

Rafailov *et al.* [?] showed that this optimization admits a closed-form solution $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(\eta r(x, y))$, which directly connects policy improvement with the reward difference between preferred and dispreferred responses. To remove the explicit reward model, DPO reformulates this closed-form expression as a supervised learning objective. Given a preference dataset $\mathcal{D} = \{(x^{(i)}, y_w^{(i)}, y_l^{(i)})\}_{i=1}^N$, where y_w and y_l denote the preferred and dispreferred responses for input x , the DPO loss is

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\eta^{-1} \log \frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \eta^{-1} \log \frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]. \quad (16)$$

This formulation directly calibrates the relative log-likelihood between preferred and dispreferred responses, providing a stable and reward-free approach for preference-based alignment of large multimodal models.

S2 Proof of Theorem 1

We provide theoretical analysis showing that, even when the reference policy π_{ref} is biased by observational shortcuts, causal preference optimization provably converges closer to the causal-optimal policy and achieves vanishing causal preference regret.

[Boundedness and realizability] The reward functions are bounded by R_{max} , the bias magnitude satisfies $\|\delta\|_\infty \leq \varepsilon$, and the learning rate η is sufficiently small so that multiplicative-weight updates remain stable.

Lemma 1 (Bias propagation under exponential tilting). *For a fixed (x) , let $q_c(y) \propto \pi_{\text{ref}}(y|x) e^{\eta r_c(x, y)}$ and $q_o(y) \propto \pi_{\text{ref}}(y|x) e^{\eta(r_c(x, y) + \delta(x, y))}$. If $\|\delta\|_\infty \leq \varepsilon$, then*

$$\text{KL}(q_o \parallel q_c) \leq 2\eta\varepsilon, \quad \|q_o - q_c\|_{\text{TV}} \leq \sqrt{\eta\varepsilon}.$$

Proof. Let $Z_c = \sum_y \pi_{\text{ref}}(y \mid x, e) e^{\eta r_c(x, y)}$ and $Z_o = \sum_y \pi_{\text{ref}}(y \mid x) e^{\eta(r_c(x, y) + \delta(x, y))}$ be the normalizing constants for q_c and q_o , respectively. By definition of the log-density ratio,

$$\log \frac{q_o(y)}{q_c(y)} = \eta \delta(x, y) + \log \frac{Z_c}{Z_o}.$$

Since δ is bounded by ε , we have $|\eta \delta(x, y)| \leq \eta \varepsilon$. To bound the partition term, note that

$$e^{-\eta \varepsilon} \leq \frac{Z_c}{Z_o} \leq e^{\eta \varepsilon} \quad \Rightarrow \quad \left| \log \frac{Z_c}{Z_o} \right| \leq \eta \varepsilon.$$

Therefore, the log-density ratio satisfies $|\log(q_o/q_c)| \leq 2\eta \varepsilon$. Integrating over q_o gives $\text{KL}(q_o \| q_c) \leq 2\eta \varepsilon$. Applying Pinsker's inequality, $\|q_o - q_c\|_{\text{TV}} \leq \sqrt{\frac{1}{2} \text{KL}(q_o \| q_c)} \leq \sqrt{\eta \varepsilon}$.

Lemma 2 (Expected one-step improvement). *Under Assumption S2, the causal preference optimization updates*

$$\pi_{t+1}(y \mid x) \propto \pi_t(y \mid x) \exp(\eta \hat{r}_c^{(t)}(x, y))$$

satisfies

$$\begin{aligned} \mathbb{E}[\text{KL}(\pi_{t+1} \| \pi_c^*) \mid \pi_t] &\leq \text{KL}(\pi_t \| \pi_c^*) - \eta \mathbb{E}_{x,e}[\text{Var}_{\pi_t}(r_c(x, \cdot))] \\ &\quad + O(\eta^2 R_{\max}^2). \end{aligned}$$

Proof. We view the policy update as a form of *mirror descent* with respect to the KL divergence. Expanding $\text{KL}(\pi_{t+1} \| \pi_c^*)$ around π_t and using the multiplicative-weight update rule, we obtain the expected change

$$\Delta_t = \mathbb{E}_x[\langle \nabla_{\pi_t} \text{KL}(\pi_t \| \pi_c^*), \pi_{t+1} - \pi_t \rangle] + \frac{1}{2} \|\pi_{t+1} - \pi_t\|^2.$$

Substituting the stochastic gradient $\hat{r}_c^{(t)} = r_c + \xi_t$ with zero-mean noise $\mathbb{E}[\xi_t] = 0$, and using the log-linear update $\pi_{t+1} \propto \pi_t \exp(\eta \hat{r}_c^{(t)})$, we obtain

$$\mathbb{E}[\Delta_t] = -\eta \mathbb{E}_x[\text{Var}_{\pi_t}(r_c(x, \cdot))] + O(\eta^2 R_{\max}^2).$$

This implies the expected KL divergence strictly decreases whenever $\text{Var}_{\pi_t}(r_c) > 0$, and thus π_t converges to π_c^* in expectation.

Theorem 1 (Convergence of causal reward). *Let $\bar{\pi}_T = \frac{1}{T} \sum_{t=1}^T \pi_t$ be the averaged iterate of causal preference optimization. Under Assumption S2,*

$$\mathcal{R}(\bar{\pi}_T) = O\left(\sqrt{\frac{\log |\mathcal{Y}|}{T}}\right), \quad \text{KL}(\bar{\pi}_T \| \pi_c^*) = O\left(\sqrt{\frac{\log |\mathcal{Y}|}{T}}\right).$$

Proof. Summing Lemma 2 over $t = 1, \dots, T$ and telescoping gives

$$\begin{aligned} \mathbb{E}[\text{KL}(\pi_{T+1} \| \pi_c^*)] &\leq \text{KL}(\pi_1 \| \pi_c^*) - \eta \sum_{t=1}^T \mathbb{E}_{x,e}[\text{Var}_{\pi_t}(r_c)] \\ &\quad + O(T \eta^2 R_{\max}^2). \end{aligned}$$

Discard the nonnegative left term to obtain $\sum_{t=1}^T \mathbb{E}_{x,e}[\text{Var}_{\pi_t}(r_c)] \leq \frac{1}{\eta} \text{KL}(\pi_1 \parallel \pi_c^*) + O(T\eta R_{\max}^2)$. For an SFT-style π_1 with full support on $|\mathcal{Y}|$ tokens, $\text{KL}(\pi_1 \parallel \pi_c^*) \leq \log |\mathcal{Y}|$. By convexity of KL and Jensen's inequality, $\mathbb{E}[\text{KL}(\bar{\pi}_T \parallel \pi_c^*)] \leq \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\text{KL}(\pi_t \parallel \pi_c^*)] = O\left(\frac{\log |\mathcal{Y}|}{T\eta} + \eta R_{\max}^2\right)$. Choosing $\eta = \sqrt{\frac{\log |\mathcal{Y}|}{T R_{\max}^2}}$ balances the terms and yields $\text{KL}(\bar{\pi}_T \parallel \pi_c^*) = O\left(\sqrt{\frac{\log |\mathcal{Y}|}{T}}\right)$.

Theorem 2 (Bias floor for observational DPO). *Let π_T^{orig} be the iterate of observational DPO trained on $r_o = r_c + \delta$, and let $\bar{\pi}_T^{\text{orig}}$ be its average. Assume there exists a set $\mathcal{A} \subseteq \mathcal{Y}$ with $\sum_{y \in \mathcal{A}} \pi_{\text{ref}}(y \mid x) \geq m > 0$ on which $|\delta(x, y)| \geq \varepsilon$. Then any limit point $\tilde{\pi}$ of $\bar{\pi}_T^{\text{orig}}$ satisfies*

$$\liminf_{T \rightarrow \infty} \text{KL}(\tilde{\pi} \parallel \pi_c^*) \geq c_0 \eta m \varepsilon, \quad \mathcal{R}(\tilde{\pi}) \geq c_1 \eta m \varepsilon,$$

for universal constants $c_0, c_1 > 0$ depending only on $R_{\max}$ and the logistic link.

Proof. Let $\pi_o^*(y) \propto \pi_{\text{ref}}(y) e^{\eta(r_c(y) + \delta(y))}$ be the optimal tilted policy under r_o at a fixed (x) , and similarly π_c^* . By the same partition-function argument as in Lemma 1,

$$\log \frac{\pi_o^*(y)}{\pi_c^*(y)} = \eta \delta(y) + \log \frac{Z_c}{Z_o}, \quad \text{with} \quad \left| \log \frac{Z_c}{Z_o} \right| \leq \eta \varepsilon.$$

Hence, for $y \in \mathcal{A}$ we have $|\log(\pi_o^*/\pi_c^*)| \geq \eta(\varepsilon - \varepsilon) = 0$ and, by one-sided averaging over $\mathcal{A}$ using $\sum_{y \in \mathcal{A}} \pi_o^*(y) \geq c m$ for some $c > 0$ (since the exponential tilting preserves a positive mass on $\mathcal{A}$), we obtain

$$\text{KL}(\pi_o^* \parallel \pi_c^*) = \sum_y \pi_o^*(y) \log \frac{\pi_o^*(y)}{\pi_c^*(y)} \geq c_0 \eta m \varepsilon.$$

As observational DPO converges to π_o^* for fixed (x) , any limit point $\tilde{\pi}$ inherits the same KL gap. Finally, apply Pinsker and Lemma 1 to transfer the KL gap to the pairwise regret lower bound.

Corollary 1 (Performance gap in causal utility). *Let $J(\pi) = \mathbb{E}_x \mathbb{E}_{y \sim \pi(\cdot \mid x)}[r_c(x, y)]$. Then for the averaged TIE-DPO iterate $\bar{\pi}_T$ and any limit point $\tilde{\pi}$ of observational DPO,*

$$J(\pi_c^*) - J(\tilde{\pi}) \geq c_1 \eta m \varepsilon - C \sqrt{\frac{\log |\mathcal{Y}|}{T}}.$$

Proof. By Hölder and Pinsker, $|J(\pi) - J(\pi')| \leq R_{\max} \|\pi - \pi'\|_{\text{TV}} \leq R_{\max} \sqrt{\frac{1}{2} \text{KL}(\pi \parallel \pi')}$. Combine the KL lower bound from Theorem 2 for $\tilde{\pi}$ with the KL upper bound of Theorem 1 for $\bar{\pi}_T$ and aggregate constants into c_1, C .

Interpretation. Causal preference optimization provably converges toward the causal-optimal policy with vanishing regret $O(T^{-1/2})$, whereas the conventional DPO objective is bounded away from the causal optimum by an irreducible *bias floor* $\Omega(\eta\varepsilon)$ due to the direct background effect.

Table 3: Medical evaluation benchmarks and statistics

Benchmark	Category	Close-ended	Open-ended	Report	Questions	Imgs
VQA-RAD	X-ray, CT, MRI	251	200	-	451	203
SLAKE	X-ray, CT, MRI	836	1,258	-	2,094	180
MIMIC-CXR	X-ray	-	-	2,195	2,195	3,681
IU-Xray	X-ray	-	-	296	296	607

S3 Data

We leverage four open-source medical vision-language datasets under two task settings: medical VQA and report generation. The training quantities for report generation denote image-report pairs, while those for VQA denote question-answer pairs. The statistics used for evaluation are summarized in the following table.

As summarized in Table 3, VQA benchmarks include SLAKE [?] (2,094 questions; 180 images) and VQA-RAD [?] (451 questions; 203 images) and spanning X-ray, CT, and MRI modalities, covering both close- and open-ended settings. Report-generation benchmarks include MIMIC-CXR [?] and IU-Xray [?], with 2,195 and 296 reports respectively (3,681 and 607 images). These statistics are used to contextualize evaluation scope and modality coverage.

- SLAKE [?] is an English-Chinese bilingual dataset comprising medical images and diverse question-answer pairs designed for Med-VQA systems.
- VQA-RAD [?] is manually curated by clinicians, featuring naturally occurring questions about radiology images with reference answers.
- IU-Xray [?] is a dataset that includes chest X-ray images and corresponding diagnostic reports.
- MIMIC-CXR [?] is a widely accessible dataset containing chest X-ray images in DICOM format along with corresponding radiology reports.

S4 Baselines

We compare our method against state-of-the-art medical vision-language models and preference optimization approaches:

STLLaVA-Med [?] is a self-training approach for medical question-answering that enhances the base LLaVA-Med model through iterative self-training techniques. It leverages pseudo-labeling and consistency regularization to improve medical visual understanding while maintaining clinical relevance.

mDPO [?] extends Direct Preference Optimization to multimodal settings by incorporating conditional visual inputs through multi-view or multi-region conditioning. This approach demonstrates how pair construction strategies and visual conditioning choices significantly affect alignment performance in medical contexts.

SPPO [?] is a self-play preference optimization method that generates preferences through self-play without requiring explicit reward modeling. It employs

a self-rewarding mechanism where the model evaluates its own outputs, simplifying the alignment process while maintaining competitive performance.

MMedPO [?] introduces clinically-aware weighting of preference samples by employing clinical relevance scores to emphasize medically significant visual-textual cues. It uses multi-agent scoring systems to identify the most clinically pertinent training examples, ensuring alignment with medical expertise rather than general visual patterns.

S5 Additional Experimental Results

S5.1 Formula of Causal Weighted DPO

In the original MMedPO framework, a weighted version of Direct Preference Optimization is introduced, where each preference pair is assigned a weight reflecting its clinical relevance. These weights are obtained through a multi-agent evaluation pipeline that estimates the degree of medical soundness for each response. Inspired by this idea, we design a causal weighting strategy that directly incorporates the causal strength of preference pairs into the DPO objective. Specifically, the causal ratio between the positive and negative samples within each pair is used as the optimization weight. The resulting causal weighted DPO objective is formulated as:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, x^*, x_c, x_c^*, y_w, y_l) \sim \mathcal{D}_f} \left[\log \sigma \left(\left(\log \frac{\pi_{\theta}(y_w | x_c) - \pi_{\theta}(y_w | x_c^*)}{\pi_{\theta}(y_l | x_c) - \pi_{\theta}(y_l | x_c^*)} \right) \left(\log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \log \frac{\pi_{\theta}(y_l | x^*)}{\pi_{\text{ref}}(y_l | x^*)} \right) \right) \right]. \quad (17)$$

S5.2 Additional Ablation Studies

Multi-round Training. In the multi-round training setup, each round initializes from the model obtained in the previous round and further refines its reasoning ability. On VQA-RAD, CaMedPO^s improves over its baseline LLaVA-Med in every round, increasing from 60.53% to 63.63% and 64.30%, while the baseline remains at 49.00%. This demonstrates that our method can continuously correct the undesired biases inherited from the initial model. For better baseline models, such as LLaVA-Med++ and Med-Gemma, a single round of training already brings substantial gains, while the subsequent training can not benefit much. Overall, the multi-round pattern shows that our approach is particularly effective at correcting models with more severe biases.

Compatibility Analysis on Other LVLMs. We further extend our preference optimization framework to two stronger medical LVLMs, LLaVA-Med++ and Med-Gemma, and use them as baselines for evaluating the general applicability of our method. On VQA-RAD, CaMedPO₊^s improves over LLaVA-Med++ from 66.52% to 71.62% after the first optimization round. Similarly, CaMedPO_G^s enhances Med-Gemma from 72.54 to 74.94 in the first round. These results demonstrate that our approach is not limited to a specific LVLM architecture and can

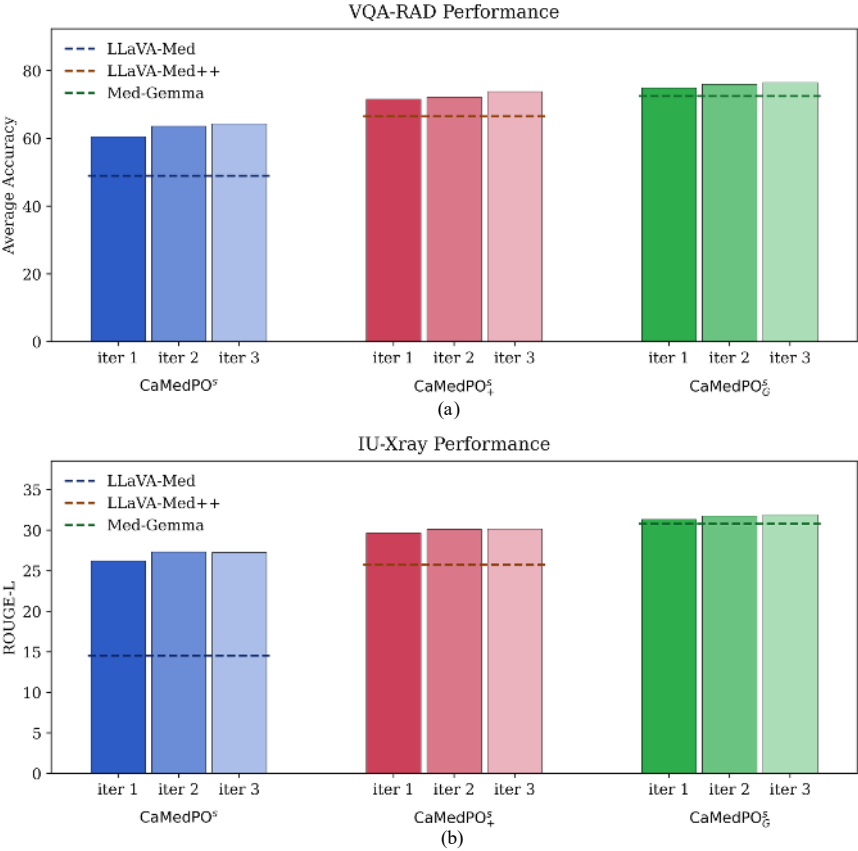


Fig. 6: Additional ablation study results on VQA-RAD and IU-XRay. ^s denotes the model enhanced with SFT. ⁺ denotes the baseline model is LLaVa-Med++, while ^G denotes the baseline model is Med-Gemma.

reliably boost performance across different baselines. Although the improvements on Med-Gemma are relatively smaller compared to LLaVA-Med, this trend is expected. Med-Gemma has undergone large-scale pretraining on diversified medical and general-domain datasets, which reduces part of the shortcut bias and leaves less room for preference-based correction. This observation suggests that as LVLMs grow in scale and diversity, further progress may require deeper incorporation of disease mechanisms and medically grounded causal structures to guide both reasoning and training. Overall, the results confirm the broad compatibility of our method and highlight its effectiveness across LVLMs with varying bias levels and training regimes.

Sample Ratio of Causal Pairs. In this section, we explore the how the ratio of the integrated causal preference loss affects the performance. In table 4, a

Table 4: Performance analysis of sample ratio of causal pairs on VQA-RAD and IU-Xray.

2*Ratio	VQA-RAD		IU-Xray		
	Open	Closed	BLEU	ROUGE-L	METEOR
0	37.00	67.73	24.00	30.13	35.17
0.5	41.00	69.72	25.56	32.81	37.32
0.75	44.00	72.11	26.43	33.76	38.05
1.0	42.50	74.90	26.23	34.13	38.25
1.25	39.00	73.70	25.12	30.83	36.07

ratio of zero corresponds to the unweighted MMedPO objective, which yields the lowest performance across both VQA-RAD and IU-Xray. Introducing moderate causal weighting leads to consistent improvements. In particular, the ratio of one achieves the best overall performance, reaching 42.50% and 74.90% on VQA-RAD open and closed questions. When the ratio becomes larger, performance begins to drop. This decline indicates that overly amplifying the causal preference term may distort the balance between the original input semantics and the causal correction, ultimately limiting the model’s reasoning capability. Overall, the results demonstrate that incorporating causal preference signals is beneficial, but the weighting must be properly calibrated to maintain stable and effective optimization.

S5.3 VQA Case Studies

Open-ended VQA Examples Figure 7 presents qualitative comparisons on open-ended medical VQA tasks. We observe that our causality-aware approach produces more clinically accurate responses by correctly identifying anatomical structures while baseline methods generate incorrect explanations based on spurious correlations.

In the presented case study, when asked “What are the white spots in the black part of the lower part of body?” in a chest X-ray image, the baseline methods incorrectly attribute the visual features to “the presence of air” with lengthy, confidence-based explanations. In contrast, our causality-aware approach correctly identifies the anatomical structures as “Lung” and “Pulmonary Bronchus” with concise, medically accurate responses. This demonstrates how our method focuses on medically relevant visual evidence while avoiding the spurious correlations that lead baseline methods to generate clinically inaccurate explanations about air presence when the actual question pertains to identifiable anatomical structures in the thoracic cavity.

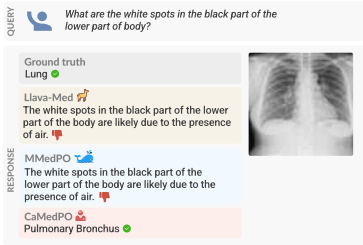


Fig. 7: Open-ended VQA case studies. Comparison of responses from baseline methods and our causality-aware approach on challenging open-ended medical questions. Green checkmarks indicate clinically accurate anatomical identifications, while red thumb down icons denote factual errors or hallucinations.

Close-ended VQA Examples Figure 8 illustrates performance on close-ended medical VQA tasks, where our causality-aware approach demonstrates superior accuracy in medical abnormality detection compared to baseline methods.

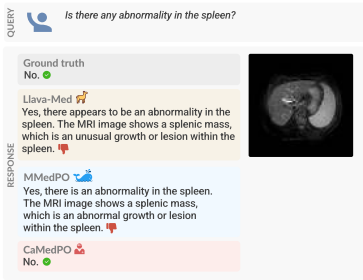


Fig. 8: Close-ended VQA case studies. Binary medical questions where our causality-aware approach correctly identifies normal anatomical structures while baseline methods incorrectly detect non-existent abnormalities. Green checkmarks indicate clinically accurate anatomical identifications, while red thumb down icons denote factual errors or hallucinations.

In the presented case study, when asked “Is there any abnormality in the spleen?” regarding an MRI image showing an axial abdominal slice, our causality-aware approach correctly answers “No” (indicating no splenic abnormality) with concise, confident responses. In contrast, baseline methods incorrectly respond “Yes” and proceed to hallucinate detailed descriptions of non-existent “splenic masses” and “abnormal growths.” This demonstrates how our causal reasoning framework prevents false positive detections that commonly plague traditional medical VQA systems, ensuring clinical decisions are based on accurate visual evidence rather than spurious pattern correlations.

S5.4 Report Generation Case Studies

Figure 9 demonstrates qualitative improvements in chest X-ray report generation, where our causality-aware approach shows superior performance in identifying complex pathological findings compared to baseline methods.

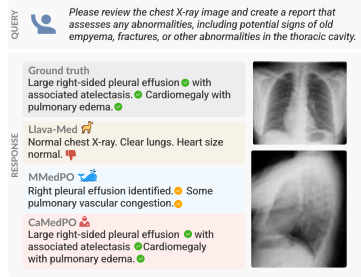


Fig.9: Medical report generation case studies. Comparative analysis of chest X-ray report quality, highlighting our method's ability to accurately identify complex pathological findings while baseline methods either miss critical findings or provide incomplete assessments. Green checkmarks indicate accurate pathological identifications, while orange checkmarks denote partial accuracy, and thumb down icons represent missed diagnoses.

In the presented case study, when instructed to “review the chest X-ray image and create a report that assesses any abnormalities, including potential signs of old empyema, fractures, or other abnormalities in the thoracic cavity,” our causality-aware approach correctly identifies the complex pathology including “Large right-sided pleural effusion with associated atelectasis. Cardiomegaly with pulmonary edema.” In contrast, baseline methods demonstrate significant limitations: one method completely misses all pathological findings and incorrectly reports “Normal chest X-ray. Clear lungs. Heart size normal,” while another provides only partial assessment by identifying “Right pleural effusion” and “some pulmonary vascular congestion” but fails to recognize the full extent of cardiomegaly and pulmonary edema. This case exemplifies how our causal reasoning framework enables comprehensive pathological assessment, ensuring that critical findings such as pleural effusions, cardiac enlargement, and pulmonary vascular changes are accurately identified and properly described in the generated medical reports.

S6 Related Works

Preference Optimization for LMs/LVLMs. Preference optimization is widely used to align large vision-language models (LVLMs) with human or model preferences [?, ?]. Early work on reinforcement learning from human feedback (RLHF) [?] trains a reward model from human comparisons and optimizes a

policy via proximal policy optimization (PPO) [?], whereas Direct Preference Optimization (DPO) [?] replaces the RL stage with a closed-form Bradley–Terry model [?] which improves efficiency and stability. Subsequent variants refine the DPO objective through margin-based, utility-based, or sampling-based strategies (KTO [?], RSO [?], ORPO [?]), while methods such as SimPO [?] and SPPO [?] further simplify alignment by removing explicit reward modeling or generating preferences via self-play. In multimodal setting, mDPO [?] extends DPO with conditional visual inputs (multi-view or multi-region), showing how pair construction and conditioning choices affect alignment. Our approach remains within the direct optimization family but introduces causal-aware sample weighting to better exploit clinically relevant visual evidence.